MUHAMMAD FAIZAN RAUF

Senior Software and AI/ML Engineer | Solution Architecture | Cloud Systems

Next.js | NestJS | Python | LLMs | Enterprise RAG | AI Agents & MCP | AWS | Docker | CI/CD

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PROFESSIONAL SUMMARY

Senior Software and AI/ML Engineer with 7+ years of experience designing distributed systems, AI-powered automation, and cloud infrastructure across FinTech, healthcare, e-commerce, and IoT. Translates business requirements into solution architecture, covering API-first design, event-driven microservices, and multi-tenant integrations. Builds AI workflows with the Anthropic API, enterprise RAG, Model Context Protocol (MCP), and multi-agent orchestration. Delivered production systems including marketplace integrations across 5+ partner APIs, read-only FHIR access to Epic and Cerner, and real-time cloud-to-device HVAC communication. Builds backends with Next.js, NestJS, and Python, deploys on AWS with Docker and CI/CD, and mentors engineers through code reviews and architecture guidance.

TECHNICAL SKILLS

Solution Architecture: System Design, Distributed Systems, Microservices, Event-Driven Architecture, API-First Design, REST APIs, GraphQL, Multi-Tenant Architecture, System Observability, Role-Based Access Control (RBAC)

AI / ML / LLM Engineering: Generative AI, Large Language Models (LLMs), Anthropic API, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, Context Window Optimization, Vector Databases, Machine Learning, Scikit-Learn, NumPy, Pandas, Data Engineering

Cloud & DevOps: Amazon Web Services (AWS), EC2, S3, CloudWatch, Docker, Docker Compose, GitHub Actions, CI/CD, Infrastructure Automation, Deployment Orchestration, Linux Server Administration, Git

Backend Development: NestJS, Node.js, Express.js, Python, FastAPI, Django, Flask, C#, ASP.NET, WebSockets, Socket.IO, Pub/Sub Messaging

Frontend Development: Next.js, React.js, TypeScript, JavaScript, Tailwind CSS, Material UI

Databases: PostgreSQL, MongoDB, MySQL, Amazon DynamoDB, Redis, SQLite

Integrations & Protocols: FHIR (Epic, Cerner), Marketplace APIs (Amazon, Etsy, Shopify, Printify, Printful), Modbus, JWT Authentication

PROFESSIONAL EXPERIENCE

Senior Software and AI/ML Engineer **Jan 2023 – Present**

PureLogics | Lahore, Pakistan

- Design solution architecture and event-driven microservices with Next.js, NestJS, and TypeScript, with clear service boundaries and data ownership.
- Build AI orchestration workflows using the Anthropic API, RAG pipelines, and MCP-connected agents for cross-platform automation, backed by Python data-processing and machine learning pipelines (NumPy, Pandas).
- Connect cloud applications to commercial HVAC equipment using Socket.IO and Modbus for real-time device communication.
- Configure AWS and Docker environments, troubleshoot releases, and optimize application performance; mentor engineers through code reviews, technical guidance, and architecture discussions.

Full Stack Engineer & DevOps **Feb 2022 – Jan 2023**

Mahir Company | Lahore, Pakistan

- Led architecture and development of the consumer web platform, admin console, and scalable MongoDB-backed APIs in collaboration with product and design teams.
- Built data-preparation workflows and API integrations supporting AI/ML features and operational automation.
- Optimized Docker environments and AWS deployment pipelines with infrastructure teams to enable continuous releases for web and API services.

Data Automation Systems Executive **Sep 2020 – Feb 2022**

NielsenIQ | Lahore, Pakistan

- Designed Python data-processing and visualization scripts for analytical software.
- Automated data ingestion and repetitive data-handling workflows, reducing manual processing effort.

Senior Software Engineer **May 2019 – Feb 2020**

The A Web Technologies | Islamabad, Pakistan

- Developed web services and MongoDB-backed data layers for client applications.

- Implemented unit tests, resolved defects, and improved application performance.

Mobile Application & Systems Engineer **Feb 2019 – May 2019**
Creativeminds | Islamabad, Pakistan

- Built Java and Kotlin mobile applications with modular client components and MongoDB-backed backend integrations.

KEY PROJECTS

HelloCustom | *E-Commerce Integration Platform*

- Designed multi-tenant, event-driven integration workflows for product personalization and order fulfillment across marketplace services.
- Integrated 5 partner APIs (Etsy, Amazon, Printify, Printful, ICP) plus Shopify.
- Implemented transaction handling and retry mechanisms for automated order-processing workflows.

NurseBrain | *Healthcare Data Integration Architecture*

- Designed read-only FHIR integrations with 2 EHR systems (Epic and Cerner), enabling real-time patient data access and reducing manual data entry.

WebShield AI | *ML-Based Phishing Detection*

- Built a Next.js/NestJS application with a Python machine learning engine and explainable website-risk analysis.
- Achieved 96.84% precision, 72.87% F1, and 58.41% recall in offline temporal evaluation on 36,910 URLs.
- Implemented GitHub Actions CI for tests, builds, PostgreSQL/Redis integration, and Docker isolation checks.

SpeakShift | *AI-Powered Video Communication Platform*

- Designed real-time video communication workflows with AI transcription, live captions, and multilingual translation.
- Built data routing and WebSocket-based group chat across connected clients.

Save Energy Systems (SES DLC) | *Cloud-Connected IoT Platform*

- Built AWS EC2-hosted IoT services using MySQL, JWT authentication, and Apache.
- Integrated Modbus and serial-port communication to connect HVAC equipment with cloud workflows.

Mahir Company | *Consumer Platform & Core API Infrastructure*

- Architected the consumer website, admin panel, and scalable MongoDB-backed APIs, streamlining data flows across product layers.
- Automated data preparation and API integration for AI/ML workflows; optimized Docker environments and AWS deployment pipelines for continuous releases.

MSQuare Market | *Blockchain Marketplace*

- Designed on-chain and off-chain wallet-service workflows for crypto-asset acquisition and management.
- Built a Next.js admin portal with role-based access control (RBAC) for requests, transactions, platforms, and users.
- Configured Docker Compose, Redis, and local DynamoDB with automated table creation and seed-data loading.

MAWA | *Real-Time Workforce Management*

- Designed task-management workflows with multilingual interfaces for distributed teams.
- Integrated audio/video calling and persistent chat within the application.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- AWS Certified Cloud Practitioner (CLF-C02) – In Progress
- Anthropic: Claude Platform 101, Claude Code 101, Claude 101

EDUCATION

Bachelor of Science in Software Engineering **Feb 2015 – Mar 2019**
COMSATS University Islamabad | GPA: 3.31

- Capstone: Wedding automation system using JavaScript, augmented reality (AR), and virtual reality (VR).
- Award: 1st Position, Open House Project Exhibition, COMSATS Project & Career Expo (Dec 2018).

LANGUAGES

English (Professional), Urdu (Native), Punjabi (Native)